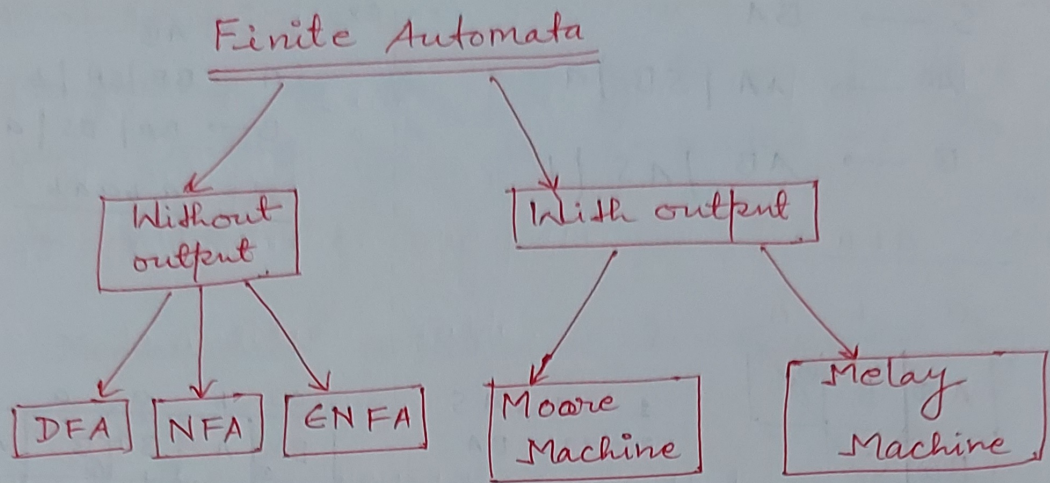




Note :-

1. Both Moore and Melay Machine are special case of DFA that is we need to define the Transition For each and every input symbol at each and every state.
2. Both Moore and Melay Machine are output producers rather than language Acceptor So no need to define any final state.

$F = \phi$

$\rightarrow$ No Final state.

$\downarrow$
Final State
3. No concept of dead state but there may be equal state.

4. Both Moore and Mealy Machine are used to ^{implement} a small count that does not require extra memory.

Moore Machine :-

The F.A. Where output is associated with state is called as Moore Machine

Moore Machine have six tuples -

$$\{ \Sigma, Q, \delta, \lambda, \Delta, q_0 \}$$

Where $\Sigma \rightarrow$ input alphabet

$Q \rightarrow$ Set of all states.

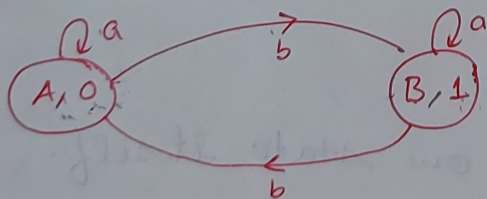
$\delta \rightarrow$ Transition Function $Q \times \Sigma \rightarrow Q$

$q_0 \rightarrow$ Initial state

$\lambda \rightarrow$ output Function.

$\Delta \rightarrow$ output alphabet.

EX



$$\Sigma = \{a, b\}$$

$$\Delta = \{0, 1\}$$

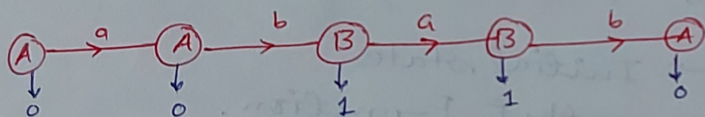
Representation of Moore Machine :-

Moore Machine can be represented in two ways

1. Transition Diagram
2. Transition Table.

Present state	Next state	output
→ A	A B	0
B	B A	1

For, $w = a b a b$ $\rightarrow$ length 4



$\lambda(abab) = 00110$ $\rightarrow$ length 5

Note :-

1. Output depends on state itself.
2. If length of input is n then length of output is $n+1$
3. Moore Machine respond for empty string ϵ that is

$$\lambda(\epsilon) = \lambda(\text{Initial state})$$

Melay Machine :- { Melay does not allow Null value, dead configuration & choices are not allowed like DFA }

$$M_{\text{elay}} = \{Q, \Sigma, \Delta, \delta, \lambda, q_0\}$$

$Q \rightarrow$ set of all states.

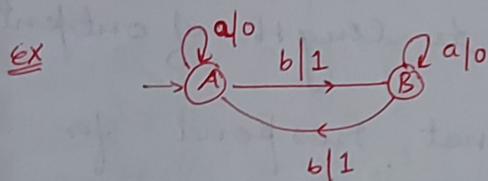
$\Sigma \rightarrow$ Input alphabets

$\Delta \rightarrow$ output alphabet

$\delta \rightarrow Q \times \Sigma \rightarrow Q$ is a Transition Function.

$\lambda \rightarrow Q \times \Sigma \rightarrow \Delta$ is output function.

$q_0 \rightarrow$ Initial state.

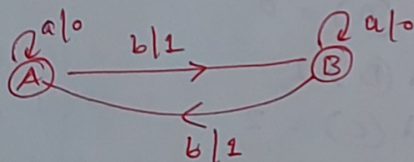


Representation of Melay Machine :-

Melay machine can be represented in two ways.

1. Transition diagram.
2. Transition Table.

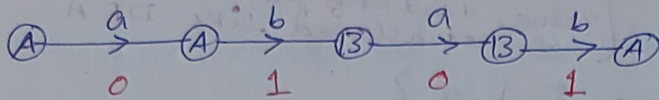
1. Transition diagram :-



2. Transition Table :-

δ	a	λ	b	λ
A	A	0	B	1
B	B	0	A	1

Now, $W = abab$:— length is 4



$\lambda(abab) = 0101 \rightarrow$ length is 4.

Note :—

1. output depends one state and input symbol.
2. Length of input equals to length of output.
3. Melay Machine can not respond for empty string 'ε'.

Conversion of Moorey to Melay Machine :-

1.)

δ	a	b	λ
$\rightarrow A$	A	C	0
B	C	B	1
C	B	A	2

Moorey Machine

$$\lambda(A) = 0$$

$$\lambda(B) = 1$$

$$\lambda(C) = 2$$

δ	a	λ	b	λ
$\rightarrow A$	A	0	C	2
B	C	2	B	1
C	B	1	A	0

Melay Machine

$$\lambda(A) = 0$$

$$\lambda(B) = 1$$

$$\lambda(C) = 2$$

2>

δ	a	b	λ
$\rightarrow q_0$	q_1	q_4	0
q_1	q_2	q_3	1
q_2	q_0	q_0	2
q_3	q_4	q_2	1
q_4	q_3	q_1	0

$$\lambda(q_0) = 0$$

$$\lambda(q_1) = 1$$

$$\lambda(q_2) = 2$$

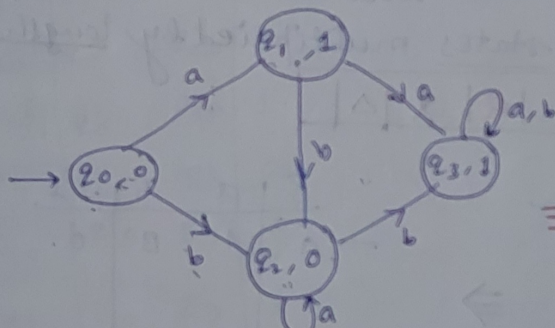
$$\lambda(q_3) = 1$$

$$\lambda(q_4) = 0$$

Moorey Machine

δ	a	λ	b	λ
$\rightarrow q_0$	q_1	1	q_4	0
q_1	q_2	2	q_3	1
q_2	q_0	0	q_0	0
q_3	q_4	0	q_2	2
q_4	q_3	1	q_1	1

Melay Machine



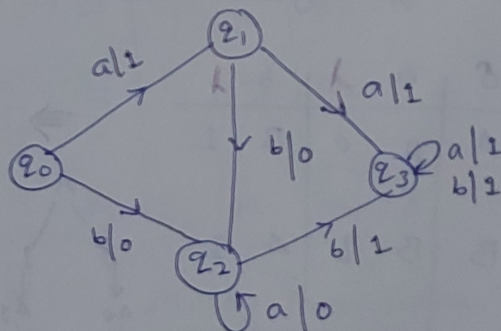
Moorey Machine

δ	a	b	λ
$\rightarrow q_0$	q_1	q_2	0
q_1	q_3	q_2	1
q_2	q_2	q_3	0
q_3	q_3	q_3	1

Transition Table of Moorey Machine

δ	a	λ	b	λ
$\rightarrow q_0$	q_1	1	q_2	0
q_1	q_3	1	q_2	0
q_2	q_2	0	q_3	1
q_3	q_3	1	q_3	1

Transition Table for Melay Machine



Melay Machine

Note 8

1. No change in the No. of state if we convert from Moore Machine to Melay Machine.

Conversion of Melay to Moore Machine

Note : 1

1. There may be change total No. of state.
2. No. of state ~~defines on~~ depends on Q, Δ .
3. Total No. of state will be less than equal to length of all states multiplied by length of output.

$$\text{Total No. of state} \leq |Q| \times |\Delta|$$

Ex

1. $\Rightarrow$

δ	a	λ	b	λ
$\rightarrow A$	A	0	B	1
B	B	1	A	0

$\Rightarrow$

B	1
A	0

Melay Machine

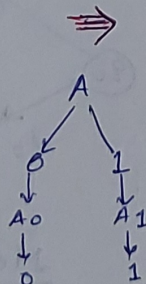
δ	a	b	λ
$\rightarrow A$	A	B	0
B	B	A	1

Moore Machine

2. $\Rightarrow$

δ	a	λ	b	λ
$\rightarrow A$	A	0	B	1
B	B	1	A	1

Melay Machine

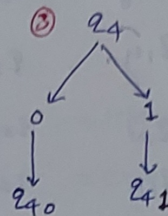
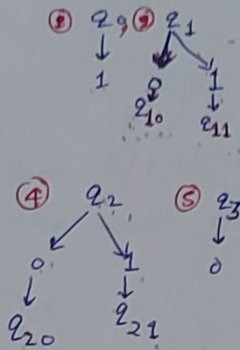


δ	a	b	λ
A_0	A_0	B	0
A_1	A_0	B	1
B	B	A_1	1

Moore Machine

3.

δ	0	1	λ
q_0	q_2	1	
q_1	q_3	0	
q_2	q_0	1	
q_3	q_4	1	
q_4	q_1	0	

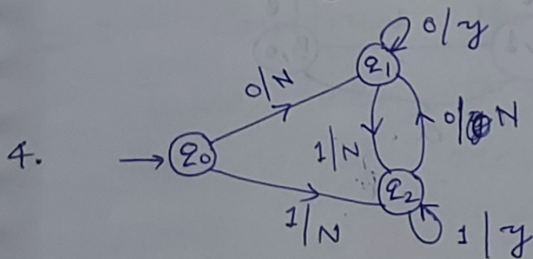


Melay Machine

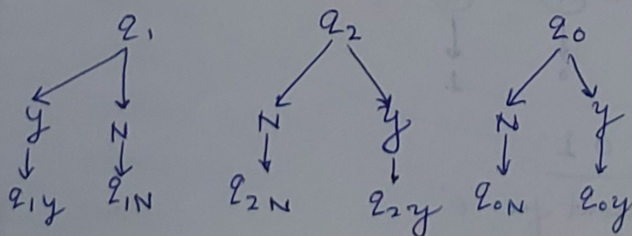


δ	0	1	λ
q_0	q_{21}	q_{20}	1
q_{10}	q_3	q_{11}	0
q_{11}	q_3	q_{11}	1
q_{20}	q_0	q_{40}	0
q_{21}	q_0	q_{40}	1
q_3	q_{41}	q_0	0
q_{40}	q_{10}	q_3	0
q_{41}	q_{10}	q_3	1

Moorey Machine



Melay diagram



δ	0	1	λ
q_0	q_1	q_2	N
q_1	q_1	q_2	N
q_2	q_1	q_2	y

Melay Transition Table

δ	0	1	λ
$\rightarrow q_{0N}$	q_{1N}	q_{2N}	N
q_{0y}	q_{1N}	q_{2N}	y
q_{1y}	q_{1y}	q_{2N}	N
q_{1N}	q_{1y}	q_{2N}	y
q_{2N}	q_{1N}	q_{2y}	N
q_{2y}	q_{1N}	q_{2y}	y

Moorey Machine

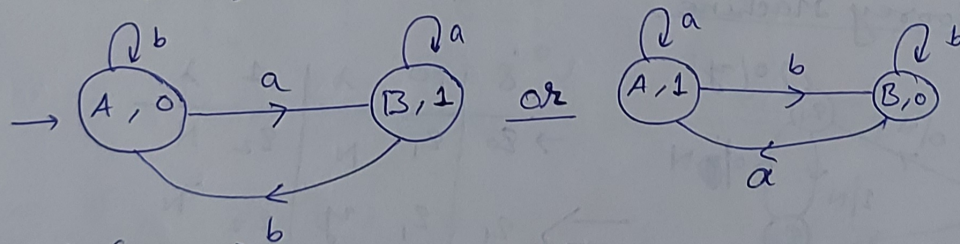
Construction of Moorey Machine :-

Question 1:-

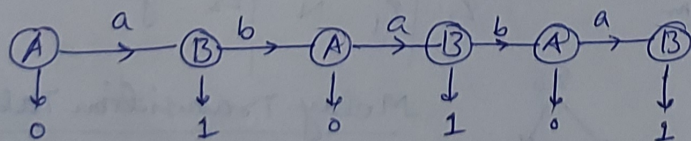
Construct the Moorey Machine that take all the string of a's and b's as input and count the no. of a's in the given I/p string.

Solution:-

$\Sigma = \{a, b\}$ Count For, a = 1 (When 'a' Accept then output is '1')
 $\Delta = \{0, 1\}$ Count For, b = 0



$\lambda(ababa)$



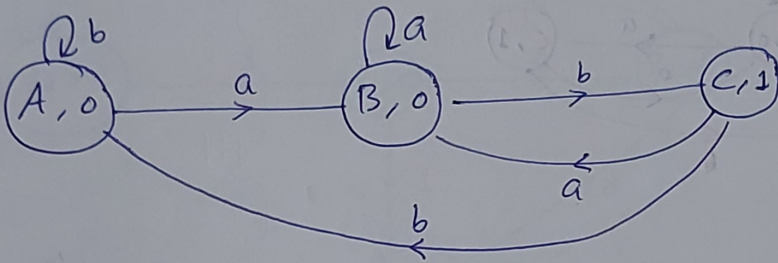
$\lambda(ababa) = 0 \underline{10101}$

↓
this represent for 'e'

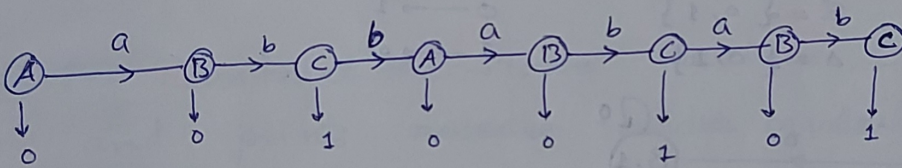
Ques 2 :- Construct the Moore Machine that takes all the input of a's and b's and count the No. of occurrence of string 'ab'

Solution :-

$ab \rightarrow 1$ (When i/p is ab then output is '1'
 $xx \rightarrow 0$ otherwise output is zero.)

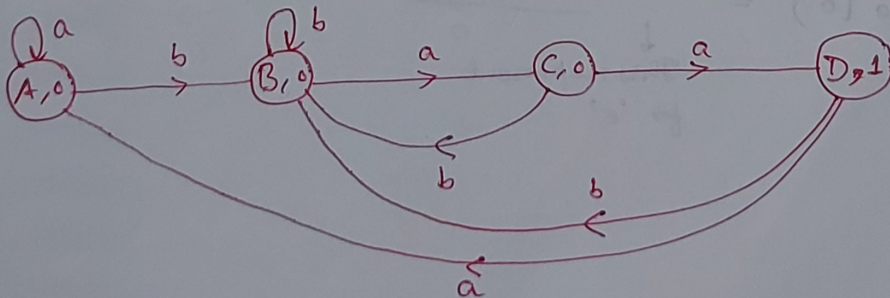


For $\lambda(abbabab)$



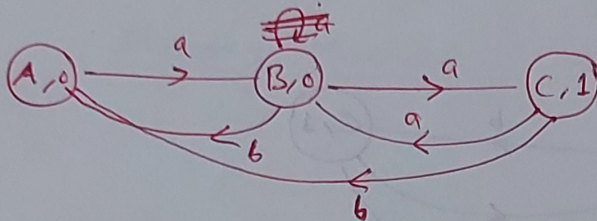
$\lambda(abbabab) = 00100101$

Ques 3 :- Construct the Moorey Machine that Take all string of a's and b's and count the no. of occurrence of string 'baa'.



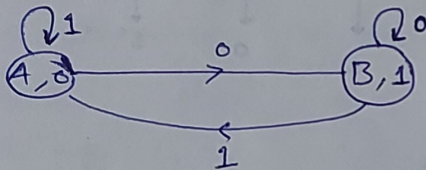
Que 4 :- Construct the Moore Machine that takes all the string of a's and b's as input and count the no. of occurrence of two consecutive a's in the given string.

Solution :-
 $aa \rightarrow 1$
 $xx \rightarrow 0$



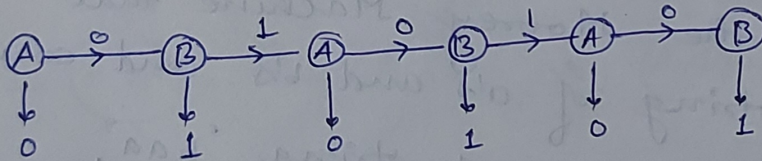
Que 5 :- Construct the moore Machine of 1's compliment in binary no.

Solution :-
 $\Sigma = \{0, 1\}$
 $\Delta = \{0, 1\}$
 $0 \rightarrow 1$
 $1 \rightarrow 0$



For,

$\lambda(01010)$



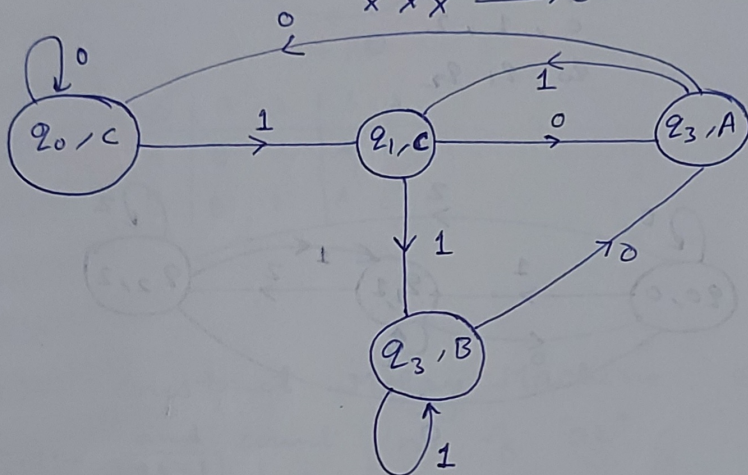
$\lambda(01010) = \underline{010101}$

↓
This represent for 'e'.

Que 6 :- Construct the Moore Machine, $\Sigma = \{0, 1\}$ and output is 'A' if input ends with 10 and output is 'B' when the input ends with '11' other wise output is 'C'

Solution :-

I/P	→	O/P
X 1 0	→	A
X 1 1	→	B
X X X	→	C



Que 7 :- Construct the Moore Machine, $\Sigma = \{0, 1\}$ and prints ~~residue~~ residue modulo 3 as output For each binary string is Treated as binary integer.

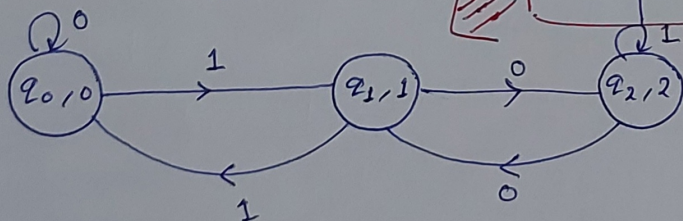
Solution

$$\Delta = \text{mod } 3$$

0, 1, 2
 q_0, q_1, q_2

δ	0	1	λ
q_0	q_0	q_1	0
q_1	q_2	q_0	1
q_2	q_1	q_2	2

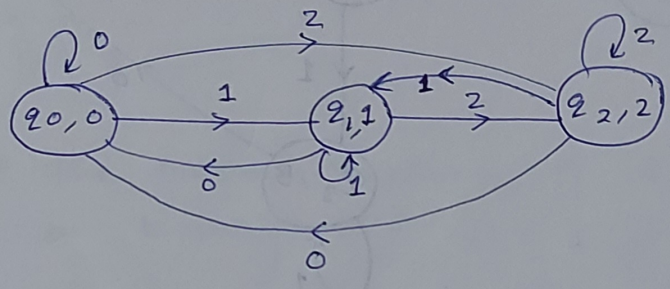
⇒ logic For binary Integer.



Ques 8 :- Construct the Moore Machine that Takes all the no. of base 3 as input and print residu modulo 3 as output.

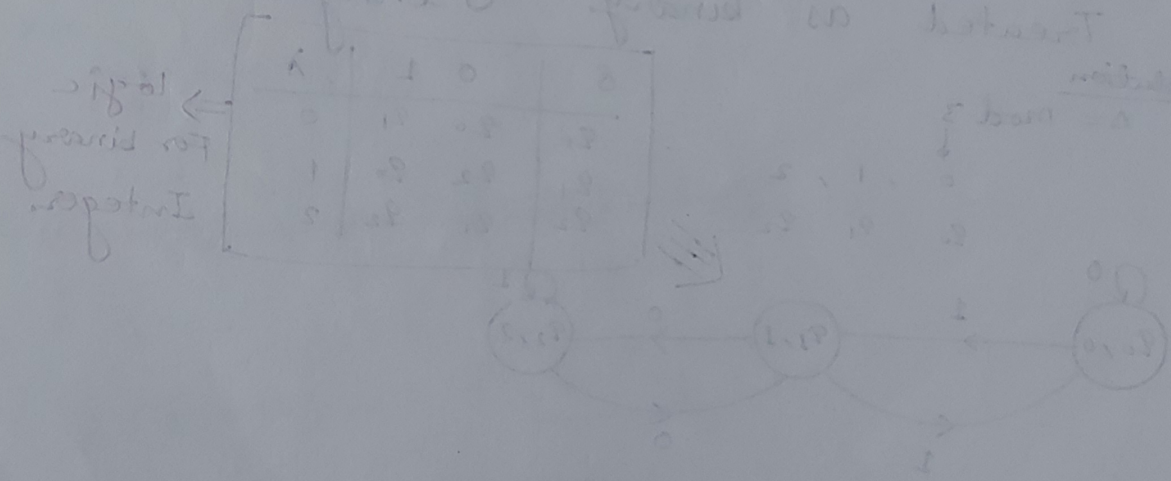
Solution:- $\Sigma = \{0, 1, 2\}$ (base 3 means 0, 1, 2 are there.)
 $\Delta = \text{mod } 3$
 $\downarrow$
 0, 1, 2
 q_0, q_1, q_2

δ	0	1	2	λ
q_0	$q_0, 0$	$q_1, 1$	$q_2, 2$	0
q_1	$q_0, 0$	$q_1, 1$	$q_2, 2$	1
q_2	$q_0, 0$	$q_1, 1$	$q_2, 2$	2



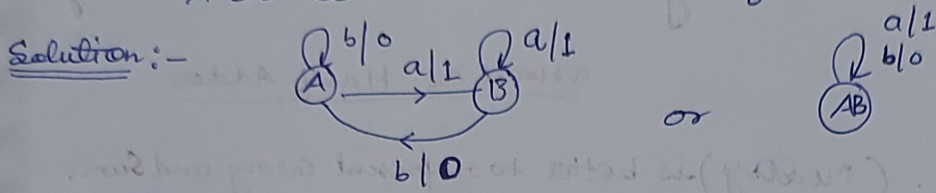
$$\lambda(012) = 2$$

$$\begin{aligned} 3^2 \cdot 3^1 \cdot 3^0 &= 27 \\ 3 + 2 = 5 / 3 = 2 &\leq \end{aligned}$$



Construction of Melay Machine :-

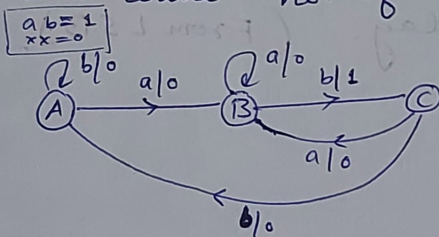
Que 1 :- Construct the Melay Machine, $\Sigma = \{a, b\}$
And count the no. of 'a'.



δ	a	λ	b	λ
A	B	1	A	0
B	B	1	A	0

Que :- Construct Melay Machine, $\Sigma = \{a, b\}$
and count no. of 'ab'.

Sol :-



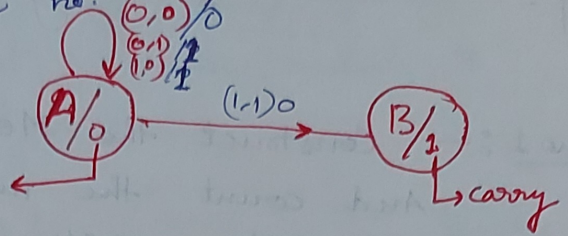
H.W.

Que 1 :- Construct Moore Machine, $\Sigma = \{a, b\}$ as input
and count the no. of occurrence of two
consecutive a's or two consecutive b's.

Que 2 :- Construct the Moore Machine that takes all the
string of a's and b's as input and output
is '1' if input ends with 'a', output is
'2' when input ends with 'ab' otherwise output is
'0'.

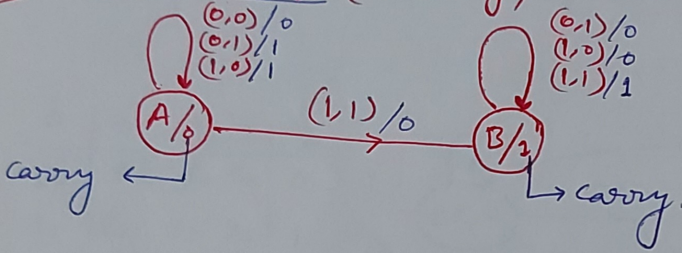
Que 3 :- Construct the ~~Moore~~ Melay Machine to compute sum of two binary no.

bit	Sum	Carry
0 0	0	0
0 1	1	0
1 0	1	0
1 1	0	1

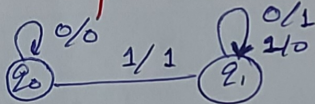


as well as Half Adder

Prob Full adder. (Melay) is better to represent carry and Sum.



Prob : 2's complement For Melay (From LSB).



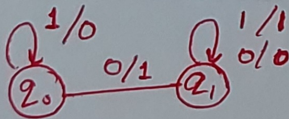
Prob :- Melay For ~~to~~ Increment by 1.

For eg.
$$\begin{array}{r} 1010 \\ + 1 \\ \hline 1011 \end{array}$$

$$\begin{array}{r} 0111 \\ + 1 \\ \hline 1000 \end{array}$$

$$\begin{array}{r} 1111 \\ + 1 \\ \hline 0000 \end{array}$$

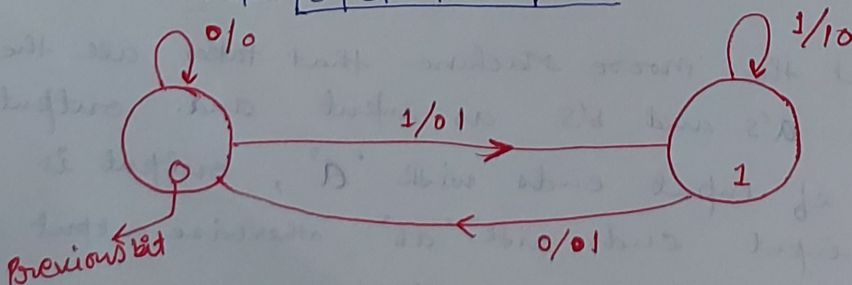
 Not possible.



Prob :-> When current bit & previous bit are added (Using MSB)

For eg.

Input	0	0 1	1 0	0 1
output	0	0 0 1	1 0 0 1	0 0 1



Melay